



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (05/08/2026 – 11/08/2026)

WEEK 8

PROJECT NAME

Ai- Based UHI mapping and mitigation

PROJECT DESCRIPTION :

**AI-POWERED SYSTEM FOR MAPPING URBAN
HEAT HOTSPOTS AND RECOMMENDING**

GROUP MEMBER :

DHAVAL AGRAWAL, OM SHAH, SANCHIT SINGH

GROUP ID :

2026G296

GROUP GUIDE :

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GROUP COORDINATOR :

SIDHDHARTH PATEL

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COLLEGE NAME :

**INSTITUTE OF ADVANCED RESEARCH,
GANDHINAGAR**

05/08/2026	• Generated and organized new training features including surface albedo, thermal resistance, and HVAC load parameters. (Dhaval Agrawal) • Updated backend data structures to support the newly introduced environmental and energy-related parameters. (Om Shah) • Absent. (Sanchit Singh)
06/08/2026	• Prepared training datasets using building physics relationships and trained the new albedo optimization model. (Dhaval Agrawal) • Extended backend APIs to handle albedo and HVAC-related model inputs and outputs. (Om Shah) • Absent. (Sanchit Singh)
07/08/2026	• Trained the HVAC prediction model and exported the optimized XGBoost models for agent integration. (Dhaval Agrawal) • Integrated new model parameters with the existing backend workflow and improved API data handling. (Om Shah) • Absent. (Sanchit Singh)
08/08/2026	• Holiday (2nd Saturday)
09/08/2026	• Holiday (Sunday)
10/08/2026	• Expanded the LangGraph agent state and updated the generative planning workflow for UHI, albedo, and HVAC optimization. (Dhaval Agrawal) • Implemented backend logic for processing optimized environmental parameters and energy-saving results. (Om Shah) • Integrated the updated AI workflow into the frontend and refined visualization of optimization results. (Sanchit Singh)
11/08/2026	• Absent due to rain. (Dhaval Agrawal) • Absent due to rain.. (Om Shah) • Absent due to rain.. (Sanchit Singh)

Next Week Plan	• Fine-tune and validate the newly trained albedo and HVAC prediction models, followed by model optimization and quantization. • Integrate the expanded LangGraph agent with the UHI, wind, albedo, and HVAC models for complete autonomous optimization. • Enhance the frontend to visualize optimized albedo, temperature reduction, HVAC savings, and AI-generated mitigation recommendations.
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REFERENCE:

https://colab.research.google.com/drive/1hYvo0OrGkDxFZ2wFijKq_b71gA51wgdN

SCREEN SHOTS:

```

print(f"[SUCCESS] Exported Model 4 ONNX Graph to: {onnx_hvac_path}")

print("\n" + "-"*60)
print("    ALL 4 TRI-DOMAIN ONNX AGENTS FULLY COMPILED & READY    ")
print("-"*60)

[INIT] Loading synthetic tri-domain dataset: /content/sankalp_tri_domain_synthetic.csv
-> Matrix loaded with 1,861,494 rows.

=====
TRAINING MODEL 3: ALBEDO SWAPPER REGRESSOR
=====
-> Model 3 RMSE : 1.0395 °C
-> Model 3 R²   : 93.34%
[SUCCESS] Exported Model 3 ONNX Graph to: /content/drive/MyDrive/AI_UHI_Project/models/sankalp_albedo_agent.onnx

=====
TRAINING MODEL 4: HVAC GRID SAVINGS PREDICTOR
=====
-> Model 4 RMSE : 0.010148 Mwh
-> Model 4 R²   : 88.32%
[SUCCESS] Exported Model 4 ONNX Graph to: /content/drive/MyDrive/AI_UHI_Project/models/sankalp_hvac_agent.onnx

=====
ALL 4 TRI-DOMAIN ONNX AGENTS FULLY COMPILED & READY
=====

1.1 Start coding or generate with AI.
1.1 Start online or generate with AI.

```

```
sankalp_tri_domain.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text Run all

# Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor

# Data Loading
X_train_alb, X_test_alb, y_train_alb, y_test_alb = train_test_split(
    X_alb, y_alb, test_size=0.2, random_state=42
)

# Model Training
model_albedo = RandomForestRegressor(
    n_estimators=300,
    max_depth=5,
    learning_rate=0.05,
    subsample=0.85,
    colsample_bytree=0.85,
    reg_lambda=3.0,
    random_state=42,
    n_jobs=-1,
    tree_method='hist'
)
model_albedo.fit(X_train_alb, y_train_alb)

# Evaluation
y_pred_alb = model_albedo.predict(X_test_alb)
mse_alb = np.sqrt(mean_squared_error(y_test_alb, y_pred_alb))
r2_alb = r2_score(y_test_alb, y_pred_alb)

print(f"Model 3 RMSE : (mse_alb*46) 'C'")
print(f"Model 3 R^2 : (r2_alb * 100).2F)%")

# Export Model 3
pk1_alb_path = os.path.join(MODEL_EXPORT_DIR, "sankalp_albedo_spent.pkl")
with open(pk1_alb_path, "wb") as f:
```



